

ESPERANZA II

Integrated Bioregenerative Food System for Mars Surface Operations

NASA Deep Space Food Challenge: Mars to Table 2026
Concept of Operations (ConOps)

Team ESPERANZA II · April 2026

Mission Parameters

15

Crew Members (Relief Crew)

500

Sols Surface Operations

3,035

kcal/person/day

45,525

kcal/day Total

Parameter	Value
Atmosphere	8.2 psi (56.5 kPa), 34% O ₂
Temperature	18–27°C (65–80°F)
Gravity	3.71 m/s ² (0.38 g)
Pre-packaged limit	≤50% of calories
Food specialists	2 only · 45 h/5 sols each (90 h combined) · worst specialist 5-sol window 81.2 h · No overtime
EVA supplement	+200 kcal/EVA-hour

Four Mission Points — NASA Operational Protocols Mapping

Verbatim traceability to Solution Summary §5 (NASA Mars to Table 2026 · Operational Protocols, 300 pts).

Mission Point	ConOps Phase	Slides	KPI Gate / Trigger
Startup	Phase 2 — Initial Surface Ops (Sols 1–60)	9–11	All 5 modules + NEXUS online $\leq$ Sol 30 · first harvest of incoming crew's own planting Sol 35 · Sol-60 gate: $\geq 53\%$ in-situ kcal, $\geq 35\%$ of it from the incoming crew's own plantings (see Slide 11)
Nominal	Phase 3 — Steady-State (Sols 60–450)	12–29	NEXUS Nominal ladder · $\geq 24,141$ kcal/sol fresh · HRV ≥ 40 ms · gut composite ≥ 70 (see Slide 18)
Contingency	Phase 5 — Emergency / Off-Nominal	30–32	Yellow → Orange → Red decision tree · 4.10 M kcal emergency rations (90 sols full) (see Slide 31)
Shutdown	Phase 4 — Hand-Over & Multi-Year Sustainment (Sols 450–500)	33 & 35	50-sol overlap · NEXUS state snapshot 3-of-3 cryptographic sign-off · no > 24 h bioregenerative pause (see Slide 35)

Phase 1 (Transit, Slides 5–8) is pre-startup; surface mission points begin at Sol 1. Coherent with Summary §5 Operational Protocols.

1. System Overview & Integration

Five integrated physical modules + the NEXUS-Loop — the kitchen manager that watches the system and advises the crew — forming a closed-loop bioregenerative food system (Solution Summary §3)



CyanoGrow

Spirulina PBRs

40 m² · 5,800 kcal/d



Crop Chambers

10 crop species

373 m² canopy · 11,366 kcal/d



FermentLab

Miso·Kefir·Tempeh·Vinegar

Value-added processing



Insect + Mushroom

Mealworms + Oyster mushroom

98 m² (48 + 50) · 6,975 kcal/d



LunaRex / MarsRem / HybridSoil

Regolith processing ·
Bioremediation · Substrate
optimization



NEXUS

The kitchen manager:
watches, advises, never
overrides
200+ sensors

12 food sources · 4 production pathways · 98.5% water recycling · 98.3% waste recycling · Integrated with habitat ECLSS

Closed-Loop Material Flow



Mars Atmosphere (95% CO₂) → CO₂ supplement: 3.1 kg/day



Plants + Spirulina
CO₂ → O₂
+15.0 kg O₂/day



Crew (15)
O₂ → CO₂
15.4 kg CO₂/day



Edible food
28.9 kg/day



Organic waste
29.7 kg/day



Insect feed
8.0 kg/day



Mushroom substrate
4.0 kg/day



Compost
12.0 kg/day

Non-recyclable waste: only 0.5 kg/day (183 kg/year) · 98.3% overall recycling rate

Caloric Budget — 500 Sols

22.76M

Total kcal required (500 sols)

≥50%

Must come from fresh production

24,141

In-situ kcal/day, crop-table nameplate (53.0%)

21,384

Earth-provisioned kcal/day (47.0%)

Source Category	kcal/day	% of Total	Role
Spirulina	5,800	12.7%	Protein, micronutrients
Staple crops (wheat, soy, potato)	10,451	23.0%	Carbs, protein, calories
Mealworms	5,600	12.3%	Protein, fat, B12
Mushrooms + vegetables + herbs	2,290	5.0%	Vitamins, variety, morale
Menu draw in excess of crop-table nameplate *	2,628	5.8%	Booked to Earth-provisioned stock — declared reserve, not a plug
Earth-provisioned items	18,756	41.2%	Caloric base, variety, cap margin — plus the 2,628 above = 21,384 (47.0%)
TOTAL	45,525	100%	

Earth-provisioned = **47.0%** ✓ (≤50% cap) · audited vs USDA FoodData Central · per-sol 32.9–47.2%, no sol over cap. * The per-ingredient audit of the 14-sol menu (Deliverable 2.2) attributes 58.8% of menu calories to in-situ ingredients. Against the *crop-table nameplate* of 24,141 kcal/sol above, ESPERANZA II conservatively books the 2,628 kcal/sol difference to Earth-provisioned stock, giving the reported **53.0% in-situ / 47.0% Earth-provisioned**. The margin is a **declared reserve, not a plug**: no crop yield has been adjusted, and 47.0% remains under the 50% cap.

Five Mission Phases

P1 Transit

P2 Setup

P3 Steady-State

P4 Hand-Over

P5 Contingency

P5 (Contingency) is not sequential: it can be entered from any phase and returns to it. Mapping to the four NASA Operational Protocol mission points: **startup** = P2, **nominal** = P3, **shutdown** = P4, **contingency** = P5.

Phase 1 — Transit to Mars

- ~7-month transfer, pre-packaged food only
- Biological maintenance & crew training

Phase 2 — Initial Surface Ops

- Relief crew handover (Sols 1–15)
- Module restart sequence (Sols 1–30)
- Own plantings ramp to full nominal by Sol 60

Phase 3 — Steady-State (Sols 60–450)

- Full production: 24,141 kcal/day fresh
- 14-sol meal rotation
- 2 specialists, 45h/5sols

Phase 4 — Hand-Over (Sols 450–500)

- 50-sol overlap with the incoming crew
- Seed & culture bank verified, NEXUS state signed 3-of-3
- System handed over **in production**: no bioregenerative pause >24 h

Phase 5 — Emergency/Contingency

- 3-tier emergency classification
- 90-sol emergency ration reserve
- Module isolation & decontamination

Phase 1: Transit to Mars

~210 Days Earth-to-Mars Transfer

Food During Transit

- 100% pre-packaged meals
- 210 days $\times$ 15 crew $\times$ 3,035 kcal = **9.56M kcal**
- 14-day menu rotation
- Supplemented by sprouting kit (radish/mung bean) for fresh micronutrients

Biological Maintenance

- Spirulina: small flask culture, weekly subculture
- Kefir grains: maintained in soy milk at 4°C
- Mealworm breeding colony: minimal population
- Seed germination tests (sample lots)
- Mushroom spawn: cold storage verification

Crew Training (En Route)

- Food specialists: 40h simulation training on system modules
- All 15 crew: 8h food safety certification refresher
- 4 backup crew: 20h cross-training for food system emergencies
- VR walkthrough of Mars food modules
- Practice HACCP monitoring procedures

Equipment Status

- All food system hardware in cargo, powered down
- Pre-existing infrastructure at Mars site: operational from previous crew
- Cargo manifest verified against spare parts inventory

Phase 2: Initial Surface Operations

Sols 1–60: System Assessment, Handover & Restart

RELIEF CREW

Arriving crew replaces existing crew. All infrastructure pre-installed & previously operational.

Sol 1 — **Inherited crop rotation feeds the crew from day one** — the outgoing crew's staggered plantings, spirulina PBR, *Tenebrio* colony and mushroom blocks all remain in production throughout handover: **24,141 kcal/sol (53.0% of demand)**, already above the $\geq 50\%$ fresh objective. **No cold start, no production gap.** The milestones below are the incoming crew's own cycle.

Sols 1–3 — NEXUS handover · Water recycling verified · System diagnostics

Sols 3–7 — CyanoGrow subculture split (inherited PBR stays online) · LunaRex/MarsRem restarted · Perchlorate baseline testing

Sols 5–10 — Crop chambers: incoming crew's first fast crops planted into the existing stagger (radish, lettuce) · LED calibration

Sols 10–15 — FermentLab re-propagated from the inherited culture bank · new *Tenebrio* cohorts set · fresh mushroom blocks inoculated

Sols 15–30 — Departing crew handover complete · All modules operational · Staple crops planted (wheat, soy, potato)

Sol 35 — **First harvest of the incoming crew's own planting: radish sprouts** (inherited production continuous since Sol 1)

Sol 60 — **Full nominal in-situ output restored: 24,141 kcal/sol (53.0%)** → Transition to steady-state

Phase 2: Food Supply Ramp-Up

Period	Earth-Provisioned	In-Situ	Key Milestones
Sols 1–15 (<i>handover</i>)	47.0% (21,384 kcal/d)	53.0% (24,141 kcal/d)	Inherited canopy, spirulina, <i>Tenebrio</i> and mushroom lines never stop producing · NEXUS handover · diagnostics · FermentLab re-propagation
Sols 15–60 (<i>ramp</i>)	47.0%	53.0%	Ferment and processing lines return to nominal · incoming crew's own plantings enter the stagger (first own harvest Sol 35)
Sol 60+	47.0% (21,384 kcal/d)	53.0% (24,141 kcal/d)	STEADY-STATE

Startup & shutdown feeding (NASA Webinar #4): no dedicated menu is required for the startup and shutdown phases, but the method of feeding the crew must be stated explicitly, and **Earth-provisioned stock is permitted** in those phases. **Startup (Sols 1–15):** ESPERANZA II is handed over *in production* — the inherited canopy, spirulina PBR, *Tenebrio* colony and mushroom blocks never stop, supplying **24,141 kcal/sol (53.0%)** from Sol 1, already above the $\geq 50\%$ fresh objective. Earth-provisioned shelf-stable stock covers the remaining **47.0%** while FermentLab cultures are re-propagated from the inherited bank and the processing line is revalidated; the transition therefore runs at the **same 47.0%** as steady state — **inside the $\leq 50\%$ cap** throughout, without invoking the startup exemption. **Shutdown / hand-over (Sols 450–500):** the departing crew leaves the same living system it received — staggered plantings, verified culture banks and a full seed inventory stay in production across the 50-sol overlap, so the incoming crew inherits continuous output and **no bioregenerative pause exceeds 24 h**.

Phase 2 Success Criteria

- All 5 modules + NEXUS operational by Sol 30
- First harvest of the incoming crew's own planting by Sol 35 (inherited output continuous since Sol 1)
- $\geq 53\%$ in-situ kcal, with $\geq 35\%$ of it from the incoming crew's own plantings — caloric production by Sol 60
- All 10 HACCP CCPs validated and monitoring active
- Water recycling rate $\geq 95\%$ (target 98.5%)
- Zero food safety incidents during ramp-up

Phase 3: Steady-State Production

Fresh Food Production Capacity (Sols 60–450)

Source	Area (m ²)	kcal/day	% Fresh	Key Nutrients
 Spirulina	40 (PBR)	5,800	24.0%	Protein (57g/100g), Fe, B-vitamins
 Wheat	150	6,027	25.0%	Carbs, protein, fiber
 Soybean	80	1,784	7.4%	Complete protein, fat, isoflavones
 Potato	80	2,640	10.9%	Carbs, Vitamin C, potassium
 Mealworms	48 (racks)	5,600	23.2%	Protein, fat, B12, Zn
 Mushrooms	50 (racks)	1,375	5.7%	Vitamin D, β-glucans, niacin
 Tomato	20	144	0.6%	Lycopene, Vitamin C
 Pepper	15	130	0.5%	Vitamin C (244mg/100g), capsaicin
 Lettuce/Kale	15	248	1.0%	Vitamin K, A, folate
 Radish	8	307	1.3%	Fast crop, micronutrients
 Herbs	5	86	0.4%	Vitamin K, flavor, morale
TOTAL	511 m²	24,141	100%	373 canopy + 40 spirulina + 48 Tenebrio + 50 mushroom

Caloric heavy-lifters: Wheat (25%) + Mealworms (23%) + Spirulina (24%) + Potato (11%) = 83% of fresh calories from 4 sources

Traceability. Crop parameters follow NASA's *Life Support Baseline Values and Assumptions Document* (Anderson, Ewert & Keener 2018, NASA/TP-2015-218570 Rev1, Tables 4-97, 4-105, 4-117 and 4-119), whose crop data the BVAD attributes to R. Wheeler (Kennedy Space Center Biomass Production Chamber) and B. Bugbee (Utah State). **The wheat yield above is deliberately conservative:** it uses the BVAD harvest index of **40%**, which Table 4-105 parameterises for cv. *Veery-10*, whereas ESPERANZA II grows **USU-Apogee** — for which Bugbee & Koerner (1997) measured harvest indices of **56% and 60%** (50–54% including roots), against 48–49% for *Veery-10* in the same trials.

2. Daily Operational Procedures

Specialist A — Crops & Systems

Time	Task
0600	NEXUS dashboard review, overnight alerts
0700	Crop harvesting & replanting (staggered)
0900	Spirulina harvest & centrifuge processing
1000	Nutrient solution: pH/EC adjust, reservoir top-up
1100	Water quality testing, perchlorate monitoring
1200	Lunch break
1300	Crop health assessment, pest/disease scouting
1400	Data logging, HACCP records, Earth report
1500	END — 9h

Specialist B — Processing & Animals

Time	Task
0600	Insect & fermentation module checks
0700	Mealworm harvest, blanch/roast/grind
0900	Mushroom harvest, new substrate blocks
1000	Food processing: tortilla baking (anti-crumb), tofu, meal prep
1130	Lunch break
1230	Fermentation management (miso, kefir, tempeh)
1400	Equipment maintenance, cleaning, sanitization
1500	END — 9h

Verified galley workload: 15.40 h/sol mean, all crew (Deliverable 2.2); worst rolling 5-sol window 80.3 h. The binding criterion is the **specialists' own 5-sol rolling window** (Rules: "45-hour workweek over five sols") — not any single sol, and not the whole-crew galley total: specialists A + B = **81.2 h of the 90 h available (90%)** · worst single specialist **40.6 h of 45** · **0 of 10 windows exceed the limit** · 2 sols rest · No overtime · NEXUS automates monitoring during rest days

Weekly Schedule (5-Sol + 2 Rest)

Sol	Specialist A (Crops/Systems)	Specialist B (Processing/Animals)
1	Wheat/soy harvest & replanting	Dough milling & starter prep (bake executed by rota) + mealworm processing
2	Spirulina bulk harvest + leafy greens	Mushroom harvest + tempeh preparation
3	Potato/root crop maintenance + water panel	Tofu production + kefir batch start
4	Full system diagnostics + perchlorate testing	Equipment PM + galley sanitation audit
5	Harvest planning + seed inventory + NEXUS calibration	Miso aging check + component batch-up for rest days
6–7	REST — System runs on NEXUS automation · Alerts only for critical deviations	

Seasonal Variation (Sol-Scale Calendar)

- **Sols 60–160:** Full ramp-up — all crops in rotation, first complete 14-sol meal cycles
- **Sols 160–350:** Peak production — surplus preserved (dried spirulina, roasted mealworm flour, grain storage)
- **Sols 350–450:** Steady production + seed stock preparation for next crew
- **Sols 450–500:** Handover preparation — document system state, culture bank verification

Staggered Planting for Continuous Harvest

Crop	Cycle (sols)	Plots	New Planting Every	Harvest Frequency
Wheat	65	7 plots × ~21 m ²	~10 sols	Every 10 sols
Soybean	90	5 plots × 16 m ²	~18 sols	Every 18 sols
Potato	90	5 plots × 16 m ²	~18 sols	Every 18 sols
Lettuce	28	4 trays × ~4 m ²	7 sols	Weekly (every 7 sols)
Radish	25	5 trays × ~1.6 m ²	5 sols	Every 5 sols
Tomato	Continuous	20 m ² permanent	N/A	Daily pick (once productive)
Pepper	Continuous	15 m ² permanent	N/A	Daily pick (once productive)
Herbs	25–35	Continuous cut-and-come	7–10 sols	Continuous
Spirulina	Continuous	40 m ² PBR	Daily	Daily
Mushrooms	14–21 per flush	10 m ² (stacked)	Weekly new blocks	Every 14–21 sols
Mealworms	60–90 life cycle	8 m ² (stacked)	Weekly cohorts	Weekly harvest

Staggering ensures no "feast-or-famine" cycles. NEXUS tracks all planting dates and generates harvest forecasts.

3. Crew Roles & Time Allocation

Food Systems Engineer (Specialist A)

- Crop production management (planting, harvest, health)
- Water & nutrient system operation
- Regolith processing oversight (LunaRex/MarsRem)
- NEXUS system administration
- HACCP Coordinator — overall food safety
- Perchlorate monitoring lead

Nutrition & Meal Prep Specialist (B)

- Food processing & preparation workflows
- Fermentation management (culture health)
- Insect & mushroom production
- Quality control testing (qPCR, mycotoxin)
- Meal planning & nutritional tracking
- Equipment maintenance lead

Time Budget (per 5-sol cycle, both combined)

Specialist task (A + B combined)	h	% of 90
In-situ ingredient processing — measured, Deliv. 2.2	28.1	31%
Crop production, harvest & replanting	22.0	24%
Monitoring & testing (HACCP, water, ClO ₄)	9.0	10%
Maintenance & repair	7.0	8%
Documentation & reporting	6.0	7%
Planning & inventory	5.0	6%
Specialist subtotal	77.1	86%
Contingency buffer	12.9	14%
TOTAL AVAILABLE	90.0	100%

Measured, not planned. Deliverable 2.2 times all 1,004 ingredient operations: **77.0 h of galley labour per 5-sol cycle**, divided as any professional kitchen divides it — **specialists turn raw biomass into food-grade components** (fermentation, tofu, tempeh, *Tenebrio*, mushroom, milling: **28.1 h**); **the crew rotation turns components into meals** (assembly, cooking, galley cleaning: **48.9 h** ≈ 45 min/person/sol × 13). Specialists carry **77.1 h of 90 (86%)**; worst 5-sol window **81.2 h (90%)**, worst specialist **40.6 h of 45, 0 of 10 exceed**. All-crew cost: **126.0 h/cycle = 18.7% of the 675 crew-hours in a 5-sol week**.

Other 13 crew: No food-system engineering duties. 4 cross-trained as emergency backups. All crew share the cooking rotation at **~45 min/person/sol**, carrying **48.9 h of the 77.0 h** galley workload — load-bearing, not a courtesy. Cook-step control points are signed off by the nutrition specialist — see Slide 28.

Human Factors & Usability

Designed so a Food Systems Engineer with general life-support training (not horticulturist) operates from Sol 1. Aligned NASA-STD-3001 Vol.2 §10 (Human-System Integration) + Hansen 2026 webinar ("process simplicity over operator skill"). **Failure tolerance:** NEXUS verifies every CCP closure vs expected sensor signature; missed/wrong-order step triggers WATCH alert in 2 min. Cognitive load capped at 6 simultaneous decisions/crew/sol (Nielsen 1994 Cognitive Walkthrough).

Ergonomics & Accessibility

- **Work-height tiers:** all routine actions (harvest, transplant, sampling, cleaning) between 0.9 m and 1.6 m above floor — within ISS-validated comfort envelope for 5th–95th percentile crew anthropometry (Mars 0.38 g)
- **Maintenance corridor:** 1.8 m clearance between rack rows · no overhead work above 1.9 m (avoid suit-induced shoulder strain in 8.2 psi suits if EVA garment used inside)
- **Tool-less quick-release** on every harvest tray, PBR tube cassette, FermentLab bioreactor lid · single-handed operation possible
- **Color-coded fluid lines** (blue=potable, green=nutrient, orange=greywater, red=brine/perchlorate) per ISO 14726 marine convention

NEXUS Operator Interface (non-expert friendly)

- **Traffic-light dashboard** (green/yellow/orange/red) — no numeric reading required for routine decisions; full data drill-down on request
- **Voice + glove-tap input** (single-hand): operator with gloves on can confirm CCP completion without removing PPE
- **Procedure prompts** auto-generated per sol (cook time, harvest order, sampling targets) — operator follows checklist, NEXUS verifies sensor signature matches expected outcome
- **Failure mode = safe:** if NEXUS down, modules continue last-known-good setpoints for ≥48 h with paper checklist fallback (printed weekly)

Tiered Training Protocol

- **Pre-launch:** Specialists A & B — 40 h simulation + 20 h hands-on at JSC Mars analog
- **Transit (~210 d):** all 15 crew — 4 h/wk system familiarization · 4 backup crew receive 20 h emergency-ops cross-training
- **On-station:** NEXUS "tutorial mode" available for any crew during first 30 sols of any rotation; ceremonial first community meal (Hand-Over slide) is also a hands-on integration drill

KPI Cadence Matrix — Operations Monitoring

Verbatim alignment with Solution Summary §5. Each cadence layer maps directly to NASA Operational Protocols rubric (300 pts). Owners and escalation triggers documented per NASA-STD-3001 Vol.2 §4.

Cadence	KPIs monitored	Owner	Trigger to escalate
Per-sol	NEXUS dashboard alerts · HRV · gut composite index · water reserve % · power kW · harvest yield vs forecast · pathogen sensor traffic-light (<i>salivary cortisol is event-driven: confirmatory sampling on WATCH, not per-sol</i>)	NEXUS auto + on-call specialist	NEXUS WATCH ladder threshold
Per-5-sol	Caloric production vs 24,141 target · perchlorate residual · pH/EC nutrient · LED PPFD drift · CCP-1 to CCP-10 + refined sub-controls CCP-3a / 4a results	Specialist A + Specialist B	>5% deviation any KPI
Per-14-sol (1 menu rotation)	Crew weight · DXA bone proxy · sodium intake (cap 2,300 mg/sol) · B12 + folate · meal acceptability survey · culture bank purity ≥95%	Medical + Specialist B	NEXUS ALERT + crew weight loss >2%
Per-rotation (500 sols)	Multi-year mass balance · equipment MTBF margin · seed-bank germination test · cryostock viability · 16S rRNA drift assay (cyanobacteria)	Both specialists + commander	drift >2% OR MTBF <500 sols margin
Per-hand-over (Sols 450–500)	Full Slide 35 checklist · documentation freeze · key rotation · ceremonial first community meal	Both crews jointly	failure-to-handover triggers Earth comm

Multi-year continuity: per-rotation KPIs feed the Multi-Year Mass Balance (≈ 91 M kcal over 4 rotations / ≈ 5.6 Earth years) — see dedicated slide. Crypto-signed NEXUS state snapshot at each hand-over (3-of-3 specialist + commander sign-off) preserves audit trail across crew turnovers.

4. Resource Management

12,000 L

Total Water in System

98.5%

Water Recycling Rate

58.1 kW

Food-system avg · 64.0 peak · 77 installed (20% margin)

98.3%

Waste Recycling Rate

Water Balance

Flow	L/day
Gross canopy transpiration	644
Condensate recovered at chamber condensers	-634.3
Recovery of transpired vapour	98.5%
— interface with the habitat, not a recovery term —	
Grey water received from habitat ECLSS	+12 (supplies the make-up)
Nutrient solution standing inventory	12,000 L (inventory, not a flow)
Net make-up required	9.7 L/sol ($644 \times 1.5\%$)

Waste Recycling Pathways

Pathway	kg/day
Composting → substrate	12.0
Insect feed (crop waste)	8.0
Mushroom substrate	4.0
Anaerobic digestion	3.7
Direct nutrient recycle (frass)	1.5
Non-recyclable	0.5

Energy Budget by Module

Module	kW avg	Scope
LunaRex (Regolith)	0.6	Crushing · leaching · electrolysis · conveying
MarsRem (Bioremediation)	2.1	Perchlorate bioreactor · UV · pumps
HybridSoil	0.4	Mixing · dosing · substrate sensors
CyanoGrow (Spirulina PBR)	4.8	PBR illumination · mixing · sparging · N ₂ fixation
FermentLab	1.9	Probiotics · fermentation · <i>Tenebrio</i> rearing
NEXUS (AI/Control)	0.9	Computing, networking, sensor fusion
TOTAL — 6 modules (SS \$3)	10.7	Duty-cycled by NEXUS

No double counting. *PBR illumination is internal to CyanoGrow and is not included in the 30.4 kW canopy figure; insect rearing is internal to FermentLab.* The 30.4 kW LED line covers **only the 373 m² of upper-tier canopy**. Loads outside these six modules — nutrient/aeration pumps, water treatment + condensate UV, mushroom chamber and substrate pasteurization, galley and task lighting — are itemized as their own lines in the power budget that follows, never inside a module.

Food-System Power Budget — 58.1 kW avg / 64.0 peak

Line item	kW avg	Basis
Modules (6, SS table)	10.7	Module table above — sums exactly to 10.7 · PBR light inside CyanoGrow
LED canopy lighting	30.4	$\Sigma(\text{DLI} \times \text{area})$ over 373 m ² canopy at 3.0 $\mu\text{mol/J}$ · peak only 33.3 kW (+9%)
Canopy HVAC (fans only , ×1.84 hypobaric correction)	5.0	8% of LED load × 1.84 — chiller work not charged (passive rejection to Mars)
Dehumidification (passive loop, pumping only)	2.2	644 L/sol latent load · 9.7 L/sol net make-up
Thermal loop + pumps	2.5	~300 m ² radiator, 40 °C loop
Nutrient / aeration pumps, 373 m ²	2.5	5–8 W/m ² of canopy
Mushroom substrate pasteurization	1.2	Batch sterilization duty
Refrigeration (4 °C food storage)	0.8	Galley + harvest cold chain
Task & corridor lighting	0.8	Crew work areas
Galley (mill, griddle, centrifuge)	0.7	~17% staggered duty · ~4 kW peak · 76 kg
Mushroom chamber	0.6	Climate + humidity control
Water treatment + condensate UV	0.4	Potable-grade condensate polish
Martian CO ₂ compressor	0.3	Atmospheric CO ₂ make-up
AVERAGE · PEAK · INSTALLED (20% margin)	58.1 · 64.0 · 77	Heat rejection: ~300 m² radiator

Power — Peak Flattening & Heat Rejection

Photoperiod staggering is a peak-flattening procedure. LED demand averages **30.4 kW** but **peaks at only 33.3 kW (+9%)**: wheat USU-Apogee runs **24/0** and acts as a constant **baseload (13.52 kW)**, while soybean and potato **tessellate opposite 12.33 h windows** (13.22 and 12.02 kW instantaneous). Scheduling photoperiods by species therefore sizes the power plant to near-average rather than to sum-of-peaks. **No spirulina-PBR or Tenebrio power is added here — already inside CyanoGrow (4.8) and FermentLab (1.9).**

Heat rejection — ~300 m², not 230. 230 m² only holds for a clean radiator against a mean sink. Martian dust deposition drives absorptivity from $\alpha \approx 0.15$ to ≈ 0.55 , and at local noon a dust-loaded radiator stops rejecting. Sizing: **~300 m² of vertical panels, 40 °C loop, oriented N-S to minimize solar view factor**, low-absorptivity coating, plus a scheduled cleaning protocol. **Radiator-cleaning EVA hours are charged to the caloric budget at 200 kcal per EVA-hour** — the light → heat → radiator → EVA → calories chain is closed explicitly.

Power — LED sole-source (no natural light): the Challenge Rules state "*Power: Not constrained*"; the team therefore sizes and justifies demand rather than capping it, and proposes surface fission power of the corresponding class. Crop light is **100% LED sole-source** at 3.0 $\mu\text{mol/J}$, sized as $\Sigma(\text{DLI} \times \text{area})$ over **373 m² of illuminated canopy = 30.4 kW**. **Why no conducted daylight:** a 373 m² translucent greenhouse loses **~46 kW by conduction** ($U \approx 1.5 \text{ W/m}^2\text{K}$, $\Delta T = 82 \text{ K}$) against only **~17 kW of optical gain** — natural light is **thermodynamically net-negative on Mars**. LED sole-source also makes production **independent of dust-storm opacity**. **Dust-storm load-shed:** lighting → leafy crops; 4.10 M kcal reserve bridges the gap. **Galley (Webinar #4 — all harvest-to-meal equipment): ~4 kW peak · 76 kg → ~0.7 kW avg** (~17% staggered duty; mill, griddle, centrifuge, grinders, press, mixer); induction (no open flame) in an O₂-trim fire-rated bay per STD-3001 Vol.2 §6.4 (SS R5).

Crop Chamber Atmosphere & Condensate Control

Independent Chamber Atmosphere (C3 photorespiration)

Wheat, soybean, potato, tomato and pepper are **all C3 species**. At the habitat's **34% O₂**, Rubisco oxygenation drives photorespiration and **net photosynthesis falls 15–25%** — every Meal Plan yield would be optimistic if the chambers shared habitat air.

- **Crop chambers hold their own atmosphere:** 56.5 kPa, ~**21% O₂**, CO₂ elevated to **1,200–1,500 ppm**
- **Airlock-separated** from the habitat, **exporting surplus O₂** to the habitat loop
- Also resolves **LED driver flammability under hyperoxia** (NASA-STD-6001): drivers sit in a **normoxic electronics bay**, DC distribution to the luminaires only

Condenser Anti-Icing

Condensing transpiration against a **–60 °C** sink yields **ice, not water**. The condensate loop is therefore actively tempered:

- **Hot-gas bypass + mixing valve** on the condenser inlet
- **Setpoint 4–10 °C** surface temperature
- **Trip if T_{loop} < 2 °C** → bypass opens, condenser sheds load before frost forms
- Protects the 644 L/sol recovery path that the 98.5% water-recycling figure depends on

5. Food Processing & Preparation

Daily Processing

 Spirulina → centrifuge → dry/press → powder/paste
1h · 12 kg/day

 Mealworms → gut purge (24h prior) → blanch/roast → grind
2h · 1.2 kg FW/day

 Fresh veg → harvest → wash (CCP-1, 50–200 ppm Cl) → serve/cook
1h

Weekly Processing

► Soft tortilla / wrap (anti-crumb format, Hansen webinar): wheat grain → mill (0.75 kW) → flour → mix → press → griddle $\geq 96^{\circ}\text{C}$ core
3h, 2×/week

 Tofu: soy → soak 12h → grind → coagulate (CaSO_4) → press
4h, 1×/week

 Kefir: soy milk + grains → ferment 24h → strain ($\text{pH} \leq 4.6$)
1h active, 1×/week

Bi-Weekly / Monthly

 Tempeh: soy → cook $\geq 95^{\circ}\text{C}$ → inoculate *R. oligosporus* → incubate 36–48h at 30–32°C
1h active, every 10 sols

 Miso: soy + koji + 12% NaCl → pack → age 60–180 sols
2h setup, checked monthly

 Vinegar: fermented sugars + mother → acetic acid $\geq 4\%$
30 min check/month

Food Preservation

- Spirulina powder: vacuum-sealed, $a_w \leq 0.30$, shelf life 180+ sols
- Mealworm flour: vacuum-sealed, $a_w \leq 0.30$, shelf life 180 sols
- Dried grain (wheat, soy): $a_w \leq 0.65$, shelf life 500+ sols
- Fermented foods: self-preserving (pH, salt, acidity)
- Fresh produce: 5–14 sols refrigerated at $\leq 4^{\circ}\text{C}$

Sample Daily Menu (14-Sol Rotation)

Day 1 Example

Meal	Items	kcal
Breakfast	Spirulina-wheat porridge, kefir, radish sprouts	780
Lunch	Potato-mushroom soup, wheat soft tortilla (anti-crumb), fresh salad w/ herbs & vinaigrette	850
Community Dinner (all 15 crew)	Soy-mealworm protein patties wrapped in soft tortilla, roasted peppers, tomato sauce, fermented chili paste (intense umami + heat per Hansen/Smith webinars)	1,005
Snack 1	Mealworm protein bar	250
Snack 2	Fresh pepper strips + miso dip	150
Total		3,035

This sol only (not the 14-sol mean — see A1): protein 98 g · fat 82 g · carbs 420 g · fibre 38 g · **Sodium** ≤2,300 mg/sol (NASA-STD-3001 Vol.1 Sec.5; miso/kéfir/tempeh batches dosified to stay under cap)

Day 8 Example — "Taco Sol"

Meal	Items	kcal
Breakfast	Wheat pancakes w/ spirulina, potato hash	820
Lunch	Miso soup, tempeh stir-fry, rice (pre-packaged)	830
Dinner	Wheat tortillas, spiced mealworm filling, tomato salsa, peppers, herbs	980
Snack 1	Kefir smoothie w/ spirulina	220
Snack 2	Roasted soy nuts	185
Total		3,035

ISS PH-04 heritage: crew "taco night" w/ chile peppers was top-rated morale event

14-sol rotation provides 78 unique meals. Pre-packaged items integrated for variety (≤50% kcal). ≤2 snacks/sol per rules.

6. Quality Control & Food Safety

HACCP-Based Food Safety Management — Mars-Adapted

Mars-Specific Hazards

- **Perchlorate (ClO_4^-):** Thyroid disruptor from regolith; bioremediation reduces 96.2% but residual risk persists
- **Closed-atmosphere pathogens:** No environmental dilution; bioaerosol accumulation
- **Radiation-induced mutation:** GCR/SPE may alter microbial pathogenicity
- **No medical evacuation:** Foodborne illness = mission-critical
- **No resupply:** Contaminated batches cannot be replaced

Monitoring Frequency

- **Continuous:** Storage temps, water turbidity (automated)
- **Daily:** Water quality, fermentation pH/temp
- **Per batch:** Crop perchlorate, insect processing temp
- **Weekly:** Pathogen testing (qPCR), surface swabs (ATP)
- **Monthly:** Urine perchlorate, environmental air sampling
- **Quarterly:** HACCP plan review, perchlorate mass balance

Detection Equipment (Mass: ~5 kg total)

- Portable qPCR (BioFire-class): Salmonella, E. coli, Listeria, Aspergillus — 45–90 min
- ATP luminometer: surface hygiene — 15 sec
- Portable ion chromatograph: perchlorate — quantitative
- Lateral flow immunoassay: mycotoxin screening
- Selective culture media: confirmation

Crew Biomonitoring

Test	Frequency	Action Level
Urine perchlorate	Monthly	>5 µg/L
Thyroid (TSH, fT4)	Every 90 sols	TSH >4.5 mIU/L
Urine iodine	Every 90 sols	<100 µg/L

Pathogen-Control Coverage

Per NASA W1 Herblet 2026 webinar: pathogen control coverage scored explicitly — 0 pts if absent. Limits and sampling cadence documented per pathogen; analytical methods cited on CCP table (next slide).

Pathogen / hazard	Critical limit	Sampling cadence	Method
<i>Listeria monocytogenes</i> (RTE foods)	n=5, c=0, absent in 25 g	per RTE batch	qPCR (BioFire-class) 45–90 min
<i>Aspergillus</i> + aflatoxin B1 (fermented foods, grains)	B1 ≤ 2 $\mu\text{g/kg}$	per fermentation cycle + grain intake	ELISA + LC-MS confirmation (CCP-4a refined sub-control)
Microcystin-LR (cyanobacteria harvest)	≤ 1 $\mu\text{g/L}$	per harvest	LC-MS quantitative (CCP-3a refined sub-control)
<i>Salmonella</i> spp.	absent in 25 g	per RTE batch	qPCR + selective culture confirmation
Total cyanotoxins (anatoxin-a, cylindrospermopsin)	≤ 2 $\mu\text{g/L}$ combined	weekly	LC-MS panel
<i>Bacillus cereus</i> (rice / starches)	$< 10^2$ CFU/g	per cooked-rice batch	selective agar plating

Hyperoxia hazard (8.2 psi, 34% O₂): accelerates thermophilic mold growth and lipid oxidation in stored foods (NASA-STD-3001 Vol.2 §6.4). Compensated by HEPA H14 + activated-carbon scrubber on FermentLab vent, dual-door air-lock to Crop Growth Chambers, and –15 Pa differential containment. **No-crumb format mandatory** (Hansen 2026 webinar) eliminates bioaerosol carriers: tortillas / wraps / gels / bars / sealed pouches — no baguette, no dry toast, no loose granola.

HACCP — 10 Core Critical Control Points

CCP	Process Step	Hazard	Critical Limit	Monitoring	Corrective Action
1	Post-harvest wash	Pathogens, surface perchlorate	Cl 50–200 ppm, ≥1 min	Each wash batch	Re-wash or cook
2	Thermal processing	Pathogens (vegetative)	Core ≥74°C, ≥15 sec	Each cooking batch	Continue cooking
3	Crop perchlorate test	Perchlorate in tissue	≤15 µg/kg (leafy), ≤70 µg/kg (other)	Every harvest batch	Quarantine → compost
4	Grain drying/storage	Mycotoxins	a _w ≤0.65, moisture ≤13%	Weekly + each batch	Re-dry; destroy if mycotoxin+
5	Fermentation pH/temp	Pathogens, bongkreikic acid	pH ≤4.6 by target time	Every 4–8h during ferment	Extend or discard batch
6	Starter culture check	Culture contamination	≥95% target morphology	Each inoculation	Discard; restart from cryobank
7	Insect thermal kill	Salmonella, B. cereus	≥80°C/5min or ≥120°C/20min	Each processing batch	Extend heating
8	Insect feed quality	Heavy metals, perchlorate	ClO ₄ ⁻ ≤50 µg/kg; Cd ≤0.5 mg/kg	Each new feed batch	Reject feed batch
9	Water treatment	All waterborne pathogens	UV ≥40 mJ/cm ² ; E. coli 0/100mL	Continuous + daily	Divert to re-treatment
10	Cold storage	Pathogen growth	≤4°C; excursion ≤2h	Continuous datalogger	Discard high-risk items

+ **refined quantitative sub-controls** per NASA W1 Herblet 2026 webinar (NOT additional CCPs — refinement of existing limits): **CCP-3a** microcystin-LR LC-MS quantitative (≤1 µg/L per harvest, within CCP-3 cyanotoxin screen); **CCP-4a** aflatoxin ELISA + LC-MS (B1 ≤2 µg/kg, within CCP-4 grain storage).

Who closes a control point when the crew rotation cooks. Routine assembly, cooking and galley cleaning are carried by the all-crew rotation (Slide 16), so three of these CCPs are executed by crew who are not food specialists: **CCP-2** (core temperature ≥74 °C, every cooked batch), **CCP-5** (fermentation pH/A_w) and **CCP-7** (*Tenebrio* thermal kill, every batch). The control is threefold: NEXUS issues a per-meal checklist and verifies each step against its expected sensor

7. Maintenance & Repair Procedures

Preventive Maintenance Schedule

Frequency	Tasks
Daily	Visual inspection, filter checks, sensor cleaning, automated NEXUS diagnostics
Weekly	Pump performance test, LED output verification, PBR integrity check
Monthly	Instrument calibration (pH, EC, thermocouple), deep clean processing equipment
Quarterly	Major system overhaul (1 module/quarter rotation), membrane replacement assessment

Repair Time Budget

7 hours per 5-sol cycle allocated to maintenance (Slide 16 specialist budget)

NEXUS predictive maintenance reduces unplanned downtime by ~40%

Spare Parts Strategy

Category	Qty	Mass (kg)
Pumps (backup units)	4	80
LED panels (replacement)	10% of installed	200
PBR tubing & connectors	20% spare	150
Sensors (pH, EC, temp, O ₂)	50% spare	30
Membranes (water filtration)	3 sets	45
Seals, gaskets, O-rings	Kit	25
3D-printable filament	50 kg	50
Misc. hardware	—	920
TOTAL		1,500

Design MTBF: >1,000 sols/component · 3D printing for non-critical parts

8. Risk Management

Risk	L	I	Rating	Mitigation	Residual
Single crop failure	M	L	MED	12-source redundancy; worst single loss 25% kcal (wheat), inside the 90-sol reserve; fast-restart crops (radish 25 sols)	LOW
Water recycling failure	L	C	HIGH	72h backup reserve (1,800L); redundant membrane units; NEXUS auto-failover	MED
Partial power outage	M	H	HIGH	Load-shedding priority list; battery backup for NEXUS; staggered photoperiods	MED
Perchlorate exceedance	M	H	HIGH	Hydroponic fallback; increased bioremediation; iodine supplementation	MED
Pathogen outbreak	L	C	HIGH	Module isolation; HACCP CCPs; emergency rations; decontamination protocol	MED
Equipment failure	M	M	MED	1,500 kg spares; 3D printing; NEXUS predictive maintenance	LOW
Specialist incapacitation	L	H	MED	4 cross-trained backup crew; NEXUS automation handles 60% of routine tasks	LOW
Insect colony collapse	L	M	MED	Separate breeding population; UV-sterilized feed; 30-sol feed reserve	LOW
Fermentation culture loss	L	M	MED	Lyophilized backup bank (50+ ampoules per culture); restart in 7–14 sols	LOW
Full module failure	VL	C	HIGH	Emergency rations (90 sols); phased restart 90–180 sols; Earth consultation	MED

L=Likelihood (VL/L/M/H), I=Impact (L/M/H/C=Critical). All risks reviewed every 180 sols or after any critical incident.

Contingency Operations

Contingency manning (12 h/sol per specialist) **exceeds the 45 h per 5-sol nominal envelope on purpose**: it is an off-nominal state bounded to ≤ 14 sols with mandatory return to nominal manning. Rules Appendix B constrains sustained maintenance workload, not emergency response.

Emergency Classification & Response

● YELLOW

Single crop/batch failure

- Quarantine affected batch
- Standard CCP corrective actions
- Adjust menu using other sources
- No emergency rations needed
- Resolve within **14 sols**

● ORANGE

Multi-crop or module-wide failure

- Emergency rations: 30–50% calories
- Fast-restart crops prioritized
- Full module decontamination
- Earth expert consultation
- Target resolution: **60 sols**

● RED

Total system failure

- 100% emergency rations deployed
- Complete module shutdown
- Root cause analysis w/ Earth
- Phased restart: 90–180 sols
- Crew medical monitoring: weekly

Emergency Ration Capacity

Item	Quantity	kcal Total	Shelf Life
Freeze-dried meal packs (500 kcal each)	5,400	2,700,000	≥ 5 years
Protein bars (250 kcal each)	5,600	1,400,000	≥ 5 years
Total capacity		4,100,000 kcal	SS \$5 verbatim
100% deployment (15 crew, 3,035 kcal/sol full)	90 sols ($4,100,000 \div 45,525 \approx 90$)		
50% supplement ($\approx 1,518$ kcal/sol)	180 sols ($4,100,000 \div 22,763 \approx 180$)		

Contingency Scenarios

Scenario A: Major Crop Module Failure

Trigger: ≥50% of crop production offline

Response:

1. Activate emergency rations (supplement 50–100%)
2. Fast-restart: radish (25 sols), lettuce (28 sols), spirulina (continuous)
3. Mealworms + mushrooms unaffected (separate module)
4. Full recovery: 60–90 sols

Caloric impact: –12,000 kcal/day → emergency rations bridge gap

Scenario B: Specialist Incapacitation

Trigger: 1 of 2 specialists unable to work

Response:

1. Activate cross-trained backup (4 crew available)
2. NEXUS handles 60% of routine monitoring
3. Reduce to essential operations only
4. Non-critical tasks deferred to next cycle

Duration: Indefinite — system designed for single-operator degraded mode

Scenario C: Water System Failure

Trigger: Primary water recycling offline

Response:

1. 72h potable reserve activated (1,800L)
2. Crop irrigation paused (plants survive 48–72h)
3. Repair priority: HIGHEST
4. Redundant membrane unit activated

Impact: Crops tolerate 2–3 sols without irrigation; full repair within 72h target

Scenario D: Contamination Event

Trigger: Pathogen confirmed in food/module (Level 2+)

Response:

1. Quarantine all product from module (72h production)
2. Module sealed — independent HEPA recirculation
3. Decontamination: peracetic acid fog + UV-C
4. 3 consecutive negative tests before restart
5. Crew medical evaluation (prior 72h consumption)

Recovery: 7–21 sols depending on scope

Multi-Year System Capability (4+ Rotations / ≈ 5.6 Earth Years)

NASA Rules p.6 — "support sustained operations for a multi-year surface mission". Each crew operates 500 sols (~ 1.4 Earth yr); system designed for **4+ sequential rotations across ≈ 5.6 Earth years** under Moon-to-Mars cadence (verbatim alignment with SS §5b).

Equipment Longevity (designed MTBF)

- LED grow array (3.0 $\mu\text{mol/J}$): $\geq 50,000$ h $\approx$ **5.7 yr** continuous (SS §5b)
- Hydroponic recirculation pumps: $\geq 35,000$ h $\approx$ 4.5 yr · **N+1** redundancy (4 backup)
- Hydroponic / PBR membranes: replace each **500-sol rotation** (≈ 18 mo, SS §5b 18–24 mo window)
- NEXUS sensor stack (200+): drift envelope $\leq 2\%$ / 5 yr · quarterly recal
- Spare-parts mass: 1,500 kg (10% of total in-habitat hardware $\sim 15,000$ kg) — replenished each 500-sol hand-over by the incoming relief crew (no mid-turn resupply; one-way mission)

Multi-Year Mass Balance

- Per rotation: 22.76 M kcal ($15 \times 500 \times 3,035$)
- Multi-rotation system load: **~ 91 M kcal** across 4 consecutive rotations ($15 \times 500 \times 3,035 \times 4$) — aligned SS §5b; architecture extensible beyond 4 rotations without redesign
- Earth-provisioned: **47.0% by calories** (menu audit 41.2%, per-sol 32.9–47.2%; the 5.8% nameplate margin booked to Earth) — audited vs **USDA FoodData Central**; 53.0% in-situ. SS §4 / §5 (R8).
- Replenishment: consumables replenished by each incoming relief crew at the 500-sol hand-over (one-way mission; no mid-turn cargo resupply) — sustains 4+ rotations / ≈ 5.6 Earth years

Crop & Culture Rotation Strategy

- Wheat / soy / potato cultivars rotated **every rotation** — disease build-up control (10–12 cultivar library on board)
- Spirulina + Anabaena starter banks refreshed **each 500-sol hand-over** from Earth cryostock — 12-mo culture viability (SS §5b)
- **16S rRNA assay** per rotation on cyanobacteria — drift $> 2\%$ triggers cryobank refresh
- Mealworm colony: **Year 3 genetic refresh** from frozen stock — inbreeding depression control
- HybridSoil substrate: post-rotation full perchlorate audit + microbial bioaugmentation

4+

Rotations · ≈ 5.6 Earth yr

5.7 yr

LED MTBF

Modular Architecture & Scalability

All 6 functional blocks (LunaRex · MarsRem · HybridSoil · CyanoGrow · FermentLab · NEXUS) are **plug-and-play modules** with standardized interfaces — enabling scale-up, scale-down, partial deployment, Moon-to-Mars reuse without redesign. **Heritage:** connector + data standards inherited from ISS US Lab racks (CCSDS, M12, 28 VDC) + NASA Mars DRA 5.0 modular surface architecture — ESPERANZA II reuses validated multi-mission interface stack, not re-invents.

Standardized Interfaces (ISO + NASA)

- **Fluidic connectors:** ISO 14726 color + Stäubli quick-disconnect (drip-free); single SKU across all modules
- **Electrical:** 28 V DC + 400 V AC backbone (NASA-STD-3008) · M12 sensor connectors · POE+ for NEXUS data nodes
- **Data:** CCSDS packets on station-internal Ethernet · open MQTT topic schema (modules publishable individually)
- **Mechanical:** ISO 668 1C-equivalent floor-pan footprint (2.44 × 2.99 m) so any module slots into a standard rack cell
- **Spares interchangeability:** 70% of replaceable parts shared across modules (pumps, valves, LED drivers, sensor heads)

Scale-Up / Scale-Down Paths

- **Scale-up to 30 crew:** duplicate CyanoGrow + FermentLab + Crop Chambers (×2 same SKU) · LunaRex + MarsRem + NEXUS retain single instance (oversized for 15 by design margin) · added mass ≈1,800 kg · added power ≈18 kW
- **Scale-down to 8 crew (Mars precursor / Moon outpost):** drop Crop Chambers tier-3 + half CyanoGrow PBRs · retain all 6 module types · 60% of baseline mass & power
- **Partial-deployment commissioning:** CyanoGrow + FermentLab alone produces 35% kcal demand from Sol 35 onward — full architecture not required for first fresh food
- **Moon outpost reuse:** identical 6-module stack runs on lunar surface (1/6 g) with LunaRex substituting Mars regolith for lunar regolith — verbatim Moon-to-Mars architecture per NASA HEOMD strategy

Evolvability

- **Open API in NEXUS** — third-party crop/culture/sensor modules pluggable via documented schema (no firmware re-flash)
- **Cultivar swap** hot-loadable between rotations: 10–12 cultivar library on board · seed bank refreshed via cargo windows

End-of-Rotation Hand-Over Protocol (Sols 450–500)

50-sol overlap window between departing & incoming crews. **Continuity guarantee:** no bioregenerative pause >24 h during transfer.

Phase 1 — Documentation Freeze (Sols 450–470)

- **NEXUS state snapshot** — verifiable hand-over artifact:
 - Full sensor archive (5 yr) ~180 GB
 - ML model weights frozen (RF LunaRex, Dirichlet-Pareto HybridSoil, MCMC-DGA, LSTM SCFA)
 - Alarm + corrective-action log (full WATCH→ALERT→CRITICAL)
 - DGA generation history (per-sol re-opt + winning formulas)
- **Triple-redundant storage:** primary SSD on-station + airlock cold backup + Earth-side DSN mirror (verified ≤Sol 480)
- **Integrity:** SHA-256 on every block · cryptographic **3-of-3 sign-off** (departing + incoming + Mission Control)
- **Recovery contingency:** 14-sol rolling buffer (always-on hot copy), ≤2% loss if snapshot corrupt

Phase 2 — Living-System Hand-Over (Sols 470–490)

- **Seed bank inventory:** 12 crops, 5-yr viability test (≥85% germination on aged lots)
- **Culture bank verified:** *Arthrospira* / *Anabaena* / *koji* / *kefir* / *T. molitor* — purity ≥95%
- **Equipment condition:** each critical component documented · MTBF margin target ≥1,000 sols
- **Medical hand-over:** per-crew thyroid + perchlorate biomonitoring history → incoming medical officer
- HACCP records + perchlorate mass-balance audit signed by both crews

Phase 3 — Authority Transfer (Sols 490–500)

- **Ceremonial first community meal** cooked jointly by both crews (Hansen webinar — community meal as social anchor)
- Final hand-over doc signed by both Food Systems Engineers
- NEXUS access keys rotated · departing crew downgraded to **read-only after Sol 500**
- Mission Control acknowledgement of successful transfer logged

3-of-3

Cryptographic sign-off

FAIL-SAFE Failure-to-handover → Earth notification + 30-sol grace period both crews on-station

NEXUS — AI Monitoring & Control

In plain terms: NEXUS is the kitchen manager. It watches the greenhouse and the galley, tells the two specialists what to harvest and cook today, and flags a problem before it becomes one — letting 2 people run what would otherwise need 6–8. It proposes; the crew decides.

Capabilities

- **200+ sensors** across all modules (pH, EC, temp, humidity, CO₂, O₂, light, flow, pressure)
- **Automated control:** nutrient dosing, photoperiod scheduling, irrigation timing, PBR circulation
- **Predictive maintenance:** ML models trained on sensor trends; alerts 7–14 sols before failure
- **Harvest forecasting:** crop growth models predict yield $\pm 10\%$
- **Anomaly detection:** flags unusual patterns (contamination, equipment degradation)

Automation Level by Task

Task	Automation
Environmental monitoring	100% automated
Nutrient dosing	95% automated
Irrigation scheduling	90% automated
Alert & alarm management	100% automated
Data logging & reporting	100% automated
Harvesting	0% (manual)
Food processing	0% (manual)
Planting/replanting	0% (manual)

NEXUS enables **rest days** — system self-manages for 48h between work cycles

No behaviour change required — the 14-sol plan meets every nutritional target in full if the crew ignores NEXUS; personalisation is opt-out. **Basis of the biometric loop:** on NASA OSDR Inspiration4 data (**n = 4, short-duration LEO analogue**), fusing all modalities separated pre- from post-flight state at LOSO 0.708 vs a null of 0.489 ± 0.107 ($p = 0.026$). The 18 cardiovascular features alone reached 0.679 — not distinguishable from the 406-feature urine panel at $n = 4$, but not significant alone. Method transfer, not result transfer. No biomarker survived correction (0 of 1112): a fused *state*, not a marker. (Puerta Angulo 2026, preprint, DOI 10.21203/rs.3.rs-10232328/v1 — detail in Appendix 3.2 §9.)

Key Differentiators

► Waste-to-Food Loops

Inedible crop biomass → mealworm feed + mushroom substrate → edible protein. Increases effective caloric yield by 25% with no additional growing area.

► 12-Source Redundancy

Single-source loss is bounded and quantified, not asserted: worst case is wheat at **25%** of fresh calories, then spirulina 24% and *Tenebrio* 23%; the other nine sources are each under 11%. No single loss exceeds the 90-sol reserve, and fast-restart crops (radish, 25 sols) rebuild the gap inside one rotation.

► NEXUS AI Enables 2-Person Operation

200+ automated sensors, predictive maintenance, and harvest forecasting allow 2 specialists to manage what would otherwise require 6–8 operators.

► ECLSS Integration

15.0 kg O₂/sol vs crew demand 12.95 (**116%**); canopy alone covers **76%**. Crew CO₂ (15.4 kg/sol) supplies **74%** of the 20.7 kg/sol fixed by crops (13.5) and spirulina (7.2); the 2.5 kg/sol balance is drawn from Martian CO₂ — an in-situ design choice, not a life-support shortfall. Relegates ECLSS electrolysis to backup duty.

► Crew Psychological Health

Fresh tomatoes, peppers, herbs, "taco night" capability, fermented foods — diet diversity score 8+ (vs. 1 for pre-packaged only). ISS PH-04 validated morale impact.

► Mars ISRU Integration

Regolith substrate + atmospheric CO₂. The closed loop recovers 98.5% of 644 L/sol transpiration → net make-up only **9.7 L/sol**, met from habitat ECLSS without dedicated ISRU water extraction.

ESPERANZA II vs. Baseline

Parameter	ISS (Current)	NASA DRA 5.0	ESPERANZA II
Crew size	6	6	15
Food source	100% Earth	100% Earth	53.0% in-situ (crop-table nameplate) + 47.0% Earth-provisioned
Crop variety	1–2	0	12 species
Growing area	0.13 m ²	0 m ²	511 m ² (373 canopy + 40 + 48 + 50)
Water recycling	93%	93%	98.5%
O ₂ from food system	~0	0	15.0 kg/day
Waste recycling	~50%	~50%	98.3%
ISRU integration	None	MOXIE only	Full (regolith, atmo, water)
Diet diversity	★	★	★★★★
Mass savings, food only, vs Earth-supply	0%	0%	54.8% (6,274 vs 13,878 kg)



Appendix

Supporting Data, References & Calculations

Reference material only — no new concepts introduced.
All concepts presented in main slides above.

A1: Nutritional Calculations

Nutrient	Daily Req. (per person)	Delivered — 14-sol mean	% of requirement
Energy (kcal)	3,035	3,060 (1,799 in-situ + 1,261 Earth)	101%
Protein (g)	56–91	123.1	135–220%
Fat (g)	67–101	94.2	93–141%
Carbohydrates (g)	380–450	423	94–111%
Fiber (g)	25–38	57.7	152–231%
Sodium (mg)	≤ 2,300 (cap)	1,853 (1,550–2,005)	81% of cap
Vitamins C · B12 · D · K	90 mg · 2.4 µg · 25 µg · 120 µg	115 · 3.2 · 25 · 220	100–183%
Iron (ceiling) · Calcium	8–10 mg · 1,000 mg	20 · 1,050	Fe 200% of ceiling · Ca 105%
Iodine (µg)	150	From the 180 kg premix ; none in situ	Urinary iodine + TSH q90 sols

Basis and its limits. Requirements are the **spaceflight** values of NASA-STD-3001 Vol.2 §7.1, not terrestrial DRIs: vitamin D is 25 µg/sol (no cutaneous synthesis inside a shielded habitat) and iron is a **ceiling**, not a target. Macronutrients and sodium are **measured** from the delivered rotation (Deliv. 2.1) and reconcile with 2.2; the energy split comes from its *Ingredient Energy Audit* sheet (58.8% / 41.2%); fat is 27.7% of energy, within AMDR. Micronutrients are **design estimates** at 70% yield derating, not an audited roll-up — stated as such so a reviewer can weight them; the premix closes any gap. **Iron is declared above ceiling at 20 mg/sol and managed:** iron-free premix, ferritin on the 90-sol panel, menu reduction next iteration. At 34% O₂ that is an oxidative-stress control, not a nutritional margin. Iodine is called out because perchlorate competitively inhibits thyroidal iodide uptake: the control is the 90-sol urinary iodine + TSH panel, which detects a real shortfall whatever the menu claims.

A2: Water Balance Detail

Crop/System	Water Use (L/m ² /day)	Area (m ²)	Gross (L/day)
Spirulina (PBR)	1.0	40	40
Wheat	1.6	150	240
Soybean	1.4	80	112
Potato	1.2	80	96
Tomato	2.0	20	40
Pepper	1.6	15	24
Lettuce/Kale	0.6	15	9
Radish	0.8	8	6
Herbs	0.6	5	3
Mushrooms	0.08	50	4
Mealworms	0.02	48	1
Substrate processing	—	—	40
Bioremediation	—	—	29
TOTAL GROSS			644
Recovery (condensate + recirculation + ECLSS, 98.5%)			–634.3
NET MAKE-UP REQUIRED			9.7 L/sol (644 × 1.5%)

A3: Atmospheric Exchange

CO₂ balance — crew-fed, Mars-topped, closed. Crew respiration (15.4 kg/sol) is the primary carbon feed, supplying 74% of the 20.7 kg CO₂/sol that crops (13.5) and spirulina (7.2) fix. Fermentation, composting and heterotrophs return a further 2.8 kg/sol; the remaining 2.5 kg/sol is drawn from abundant Martian CO₂ (3.1 kg/sol makeup, 0.6 kg/sol buffered). *Using in-situ CO₂ to feed photosynthesis is a design choice, not a life-support shortfall.*

O₂ coupling to ECLSS. The crop canopy alone releases **9.82 kg O₂/sol = 76% of the crew's metabolic demand (12.95 kg/sol)**, before counting the spirulina PBR. The food system is therefore a primary atmospheric revitalization asset, not just a food source.

Source/Sink	O ₂ (kg/day)	CO ₂ (kg/day)
Crop photosynthesis (373 m ² canopy)	+9.82	−13.5 (consumed)
Spirulina photosynthesis (40 m ² PBR, 7,200 L)	+5.2	−7.2 (consumed)
Crew respiration (15 people)	−12.95	+15.4
Fermentation	—	+1.5
Composting	—	+0.8
Insects/mushrooms (heterotrophs)	−0.4	+0.5
Mars atmosphere extraction	—	+3.1 (makeup — see note)
NET	+1.7 surplus	+0.6 surplus (21.3 produced / 20.7 fixed)

The food system (crops + spirulina) releases 15.02 kg O₂/sol against a crew demand of 12.95 — **116% coverage** (canopy alone 76%), leaving a net **+1.7 kg/sol surplus** after heterotroph consumption. Habitat ECLSS electrolysis is therefore relegated to backup and transient duty. *MarsRem's perchlorate reduction does liberate O₂ stoichiometrically, but it is claimed here only as a detoxification process: the regolith throughput needed to make it an atmospheric contributor is not part of this architecture.*

A4: 500-Sol Crop Calendar (Simplified)

Planting and inoculation dates below are the **incoming crew's own cycle**, laid over the inherited stagger. The lines received from the departing crew stay in production continuously from Sol 1 (see Phase 2).

Source	Sols 1–30	Sols 30–60	Sols 60–160	Sols 160–350	Sols 350–500
Spirulina	Inherited PBR in production · subculture split Sol 3	Ramp up	Full production	Full production	Full + surplus dry
Radish	Plant Sol 5	Harvest Sol 30+	Continuous (5-sol cycle)	Continuous	Continuous
Lettuce/Kale	Plant Sol 8	Harvest Sol 36+	Continuous (7-sol cycle)	Continuous	Continuous
Mushrooms	Inherited blocks flushing · new blocks Sol 15	First flush Sol 35	Continuous (14-sol flush)	Continuous	Continuous
Wheat	Plant Sol 15	Growing	First harvest Sol 80	Staggered (10-sol new plots)	Maintain + seed stock
Soybean	Plant Sol 20	Growing	First harvest Sol 110	Staggered (18-sol plots)	Maintain + seed stock
Potato	Plant Sol 20	Growing	First harvest Sol 110	Staggered (18-sol plots)	Maintain
Tomato	Plant Sol 10	Growing	First fruit Sol 90	Continuous pick	Continuous
Pepper	Plant Sol 10	Growing	First fruit Sol 100	Continuous pick	Continuous
Herbs	Plant Sol 8	First cut Sol 33	Continuous cut-and-come	Continuous	Continuous
Mealworms	Inherited colony harvesting · new cohorts Sol 10	Growing	First harvest Sol 70	Weekly cohort harvests	Maintain breeding pop.

A5: Mass Budget Summary

Category	Mass (kg)	Notes
Core modules (LunaRex, MarsRem, HybridSoil, CyanoGrow, FermentLab, NEXUS)	2,085	Aligned with SS §3
Ancillary in-habitat hardware	~12,865	LED arrays, hydroponic plumbing, racks, processing equipment, controllers
Biological starters (seeds, cultures, colonies)	205	300% seed redundancy included
Nutrient salts (500-sol supply)	700	Scaled to 500-sol supply (replenished each rotation)
Emergency food reserves	1,500	90 sols / 4.10 M kcal. Mass derived: $5,400 \times 0.13 + 5,600 \times 0.06$ kg +35% packaging $\approx 1,500$ kg (SS §5)
Spare parts & maintenance	1,500	10% of total in-habitat hardware (~15,000 kg)
Water (initial system charge)	8,000	Seeding the closed loop
Growth media & substrates	600	Perlite, vermiculite
Earth-provisioned food (500 sols)	6,274	47.0% of 45,525 kcal/day $\times$ 500 sols
TOTAL IMPORTED	~33,729	incl. 8,000 kg initial water charge, excluded from SS §4 Earth food budget (33,729 – 8,000 = 25,729)

Comparison: Fully Earth-supplied food for 15 crew $\times$ 500 sols = 1.8 kg/person/day $\times$ 15 $\times$ 514 days $\approx$ 13,878 kg (food only). ESPERANZA II total system mass is higher but includes all infrastructure for ongoing self-sufficiency and next-crew handover.

A6: HACCP Decision Tree (Codex Alimentarius Adapted)

Logic applied to identify each Critical Control Point (CCP-1 to CCP-10) plus refined sub-controls CCP-3a (microcystin LC-MS) and CCP-4a (aflatoxin ELISA). Codex Alimentarius CAC/RCP 1-1969 Rev. 2020 + FDA 21 CFR Part 120/123.

Q1. Do preventive control measures exist for the identified hazard at this step?

▼ YES → Q2 · NO → modify step until control exists

Q2. Is the step specifically designed to eliminate or reduce the hazard to an acceptable level?

▼ YES → **CCP confirmed** · NO → Q3

Q3. Could contamination occur or increase to unacceptable levels at this step?

▼ YES → Q4 · NO → not a CCP

Q4. Will a subsequent step eliminate or reduce the hazard to an acceptable level?

▼ YES → not a CCP · NO → **CCP confirmed**

Mars-specific adaptation: the 10 core CCPs plus sub-controls CCP-3a and CCP-4a (12 control points in total) are additionally screened against (a) perchlorate ingestion, (b) closed-atmosphere pathogen amplification (recirculation), (c) radiation-induced microbial mutation (ferment cultures, cyanobacteria) — adds 3 supplementary screening criteria beyond Codex.

A7: Spare Parts Inventory (1,500 kg — 10% of total in-habitat hardware ~15,000 kg)

Component	Qty	Mass (kg)	Criticality	MTBF target
Hydroponic recirculation pumps	4	120	CRITICAL	≥35,000 h
LED panels (full spectrum + far-red)	~10% deployed	180	CRITICAL	≥50,000 h
PBR membrane sets (CyanoGrow)	3 full sets	95	CRITICAL	~18 mo (replace/rotation)
NEXUS sensors (pH/EC/temp/CO ₂ /O ₂ /PPFD)	40 spares	30	HIGH	quarterly recal
Centrifuge spirulina harvest unit	1 backup	85	HIGH	≥10,000 cycles
Mealworm trays + frass screens	20 sets	110	MEDIUM	≥5 yr
FermentLab vessels + air-lock seals	8 vessels	140	HIGH	≥3 yr
HEPA H14 filter cartridges	6 cartridges	45	HIGH	annual replace
Activated-carbon scrubbers	4 cartridges	38	MEDIUM	annual replace
UV-C water treatment lamps	6 lamps	22	CRITICAL	≥9,000 h
Hydroponic salts + trace minerals (buffer)	—	180	HIGH	5 yr shelf-life
3D-printable polymer feedstock (PLA + PET)	—	250	MEDIUM	indefinite
Tubing, valves, fittings, O-rings	assorted	180	MEDIUM	≥7 yr
Electronics modules + cabling	assorted	25	HIGH	≥10 yr
TOTAL SPARE PARTS	—	~1,500	—	5-yr horizon

3D-printing capability: non-critical parts (clips, brackets, tray inserts, sensor housings, valve handles) printed on-station from PLA/PET feedstock. Reduces stockpile mass ~15%.

A8: System Model Parameters

Key Inputs

Parameter	Value
Crew size	15
Mission duration (rotation)	500 sols (513.75 d)
Multi-rotation horizon	≈5.6 Earth yr (4+ rotations)
kcal/crew/sol	3,035
EVA supplement	+200 kcal/EVA-h
Habitat pressure	8.2 psi (56.5 kPa)
Habitat O ₂	34%
Habitat temp	18–27 °C
Mars gravity	3.71 m/s ²
Photoperiod (per species)	Wheat USU-Apogee 24/0 · Soybean 12.33/12.33 · Potato 12.33/12.33 · leafy & fruiting crops 16/8
LED efficacy	3.0 μmol/J
Lighting mode	LED sole-source (no conducted daylight)
Illuminated canopy	373 m ² → 30.4 kW LED
Growing area total	511 m ² = 373 canopy + 40 spirulina + 48 Tenebrio + 50 mushroom
Pressurised volume (food system)	1,058 m³ total (per-module breakdown in Design Layout)

All values above are generated by the **ESPERANZA II system model** — a Python implementation of the five coupled modules (~1,900 lines) an internal engineering model, run standalone for the results reported here. Closed-loop runs span the full 500-sol rotation; off-nominal contingencies validated: power blackouts (Scenario E), water restriction (Scenario C), EVA peaks, dust storms ≥14 sols. Model and complete parameter set available to the Challenge Administrators on request.

Validated Outputs vs NASA Reference

KPI	ESPERANZA II	NASA ref
Fresh kcal/sol	24,141	≥22,762 (50%)
O ₂ produced kg/sol (system total: crops + spirulina)	15.0	~12.0 (DRA 5.0)
CO ₂ uptake kg/sol (system total)	20.7	~16.5 (DRA 5.0)
Water recycle %	98.5%	93% (ISS)
Waste recycle %	98.3%	~50% (ISS)
Earth provisioned % (by calories)	47.0% (menu audit 41.2% + 5.8% nameplate margin)	≤50% by calories (Rules p.17, Table 3)
Food-system power (kW)	58.1 avg · 64.0 peak · 77 installed	Rules: <i>"Power: Not constrained"</i>
Heat rejection (radiator)	~300 m ² · 40 °C loop	Dust-loaded α ≈ 0.55 case

ML Models (NASA datasets)

- **RandomForest** — LunaRex regolith yield
- **Gaussian Process** — LunaRex spatial mineral interpolation
- **Dirichlet-Pareto** — HybridSoil 6-objective substrate blend
- **DGA (MCOMM)** — per-sol meal re-optimization
- **LSTM** — SCFA / gut composite forecast

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ESPERANZA II

Feeding 15 on Mars — 500 Sols of Self-Sufficiency

12

Food Sources

53.0%

In-Situ Calories

98.5%

Water Recycling

2

Specialists · 45h/5sols

Closed-loop · AI-enabled · Mars-ISRU integrated · HACCP-certified · Crew-centered

Team ESPERANZA II · NASA Mars to Table Challenge 2026